Jackson R. Strand

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Early career professional and PhD student in Ecology & Environmental Sciences at Montana State University, with a background in entomology and agroecology. My work integrates chemical ecology, host-herbivore interactions, and invasive species management, currently focusing on the biological control of yellow and Dalmatian Toadflax. Experienced in volatile analysis, life table studies, and community ecology, with a strong record of publications, presentations, and scientific leadership. More information is available at www.jackson-strand.com

EDUCATION

PhD **Montana State University** 2025 - present
Ecology & Environmental Science
Graduate Committee: Dr. Robert Peterson (co-chair), Dr. David Weaver (co-chair), Dr. Sharlene Sing, Dr. William Wetzel

MSc **Montana State University** 2021 - 2024
Entomology
Graduate Committee: Dr. David Weaver (co-chair), Dr. Tracy Sterling (co-chair), Dr. Robert Peterson
Graduate Certificate in Applied Statistics

BSc **University of Minnesota - Twin Cities** 2013 - 2017
Biology
Minor: Marine Biology

PUBLICATIONS

1. Strand JR, Burkle LA, Cook D, Gardner DR, Keith B, Wallander R, Weaver DK, Sterling TM. 2026. Volatile insensitivity and localized alkaloids support commensalism in the *Oxytropis sericea–Alternaria oxytropis* association. Journal of Chemical Ecology. (Accepted)
2. Peterson RKD, Kennedy TE, Strand JR. 2026. Risk Determination for Non-Target Arthropods from Outdoor Space Applications of Mosquito Adulticides. Journal of Medical Entomology. 63(3). DOI: 10.1093/jme/tjag090
3. Strand JR, Peterson RKD, Sterling TM, Weaver DK. 2025. Sensory and behavioral responses of braconid parasitoids to changes in volatile emissions induced by wheat stem sawfly (Hymenoptera: Cephidae) larval feeding in winter wheat and smooth brome. Journal of Insect Science. 25(2). DOI: 10.1093/jisesa/ieaf016
4. Strand JR, Peterson RKD, Sterling TM, Weaver DK. 2024. Agroecological importance of smooth brome in managing wheat stem sawfly (Hymenoptera: Cephidae) via associated braconid parasitoids. Journal of Economic Entomology. 117(6): 2344–2354. DOI: 10.1093/jee/toae246

HONORS & AWARDS

Crawford Wildlife Habitat Graduate Fellowship ($10,000) 2026
Montana State University

Lloyd A. Anstutz Agricultural Studies Scholarship ($950) 2026
Montana State University

Extreme Biofilms NRT Program SEED Grant ($1,500) 2025
Extreme Biofilms NRT, Montana State University

Angus and Anne McMillan Family Agriculture Scholarship ($1,200) 2025
Montana State University

Open Access Author Fund ($1,736) 2025
Montana State University Library

Funded the publication of *Strand et al. 2025*

NAISMA 2025 Student Scholarship 2025
National Forum on Biological Control, Annapolis, MD

Graduate Land Grant Excellence Award 2024
The Graduate School, Montana State University

Open Access Author Fund ($3,296) 2023
Montana State University Library
Funded the publication of *Strand et al. 2024*

President's Prize - 2nd Place Speaker 2023
Entomology 2023, National Harbor, MD
Entomological Society of America

Student Scholar Travel Award ($500) 2023
College of Agriculture, Montana State University

1st Place Speaker 2023
2023 LRES GSO Student Research Colloquium
LRES Graduate Student Organization, Montana State University

Montana Grains Foundation Scholarship ($2,500) 2022
Montana Grain Growers Association

Graduate Professional Advancement Grant ($500) 2022
The Graduate School, Montana State University

PRESENTATIONS

Invited presentations at scientific meetings

1. Strand JR. 2026. From Release to Persistence: How exclosures may enhance biocontrol success. Montana Weed Control Association Conference. Billings, MT.

Contributed presentations at scientific meetings

1. SE Sing, Strand JR, Peterson RKD, Weaver DK. 2025. Understanding Host Preference in *Mecinus janthinus* and *M. janthiniformis*: Implications of Hybridization for Effective Biological Control. National Forum on Biological Control Conference. Annapolis, MD.
2. Strand JR, Sterling TM, Peterson RKD, Weaver DK. 2025. Agroecological Importance of Smooth Brome in Hosting Braconid Parasitoids Associated with the Insect Pest Wheat Stem Sawfly. Annual Meeting of the Western Society of Weed Science. Seattle, WA. (poster)
3. Strand JR, Peterson RKD, Sterling TM, Weaver DK. 2024. Agroecological importance of smooth brome in managing wheat stem sawfly (Hymenoptera: Cephidae) via its associated braconid parasitoids. Session: Grad Competition, Plant-Insect Ecosystems: IPM – General and Field Crops. Annual Meeting of the Entomological Society of America, Phoenix, AZ.
4. Strand JR, Weaver DK, Sterling TM, Peterson RKD. 2024. The role of chemical ecology and agroecological importance of smooth brome in biological control of the wheat stem sawfly by native parasitoids. Masters of Entomology Defense. Montana State University, Bozeman, MT.
5. Sing SE, Ward SM, Peterson RKD, Strand JR, Friesenhahn E, Holland M, Hubbard CE, Randall CB, Weaver DK. 2024. Biological control of hybrid toadflax: insights from chemical ecology. National Forum on Biological Control Conference. Annapolis, MD.
6. Strand JR. 2024. How smooth brome and parasitoids can aid in the biological control of wheat stem sawfly. Montana Wheat and Barley Committee - March Madness. Montana State University, Bozeman, MT.
7. Strand JR, Weaver DK, Sterling TM, Peterson RKD. 2023. Using organic volatile compounds to assess the relationship between wheat stem sawfly, parasitoids, and smooth brome. Session: Grad Competition, Plant-Insect Ecosystems: Chemical Ecology 2. Annual Meeting of the Entomological Society of America, National Harbor, MD. **Second place in section**.
8. Strand JR, Weaver DK, Sterling TM, Peterson RKD. 2023. The importance of smooth brome and parasitoids in wheat stem sawfly biocontrol. 2023 Land Resources and Environmental Sciences Gradu-

ate Student Organization Research Colloquium. Montana State University, Bozeman, MT. **First place overall**.

9. Strand JR, Weaver DK, Sterling TM, Peterson RKD. 2022. Using organic volatile compounds to assess the relationship between wheat stem sawfly, parasitoids, and smooth brome. Session: Grad Competition, Plant-Insect Ecosystems: Chemical Ecology 1. Annual Meeting of the Entomological Society of America. Vancouver, BC, Canada.
10. Strand JR, Weaver DK, Sterling TM, Peterson RKD. 2022. Importance of parasitoids and smooth brome for integrated pest management of wheat stem sawfly. 2022 Land Resources and Environmental Sciences Graduate Student Organization Research Colloquium. Montana State University, Bozeman, MT.
11. Strand JR, Weaver DK, Sterling TM, Peterson RKD. 2021. Importance of parasitoids and smooth brome for integrated pest management of wheat stem sawfly. Session: Plant Insect-Ecosystems: IPM – General. Annual Meeting of the Entomological Society of America, Denver, CO.

Invited seminars

1. Strand JR. Can Bugs Help Us? ENSC 210: Role of Plants in the Environment. Montana State University. Instructor: Will Wetzel. March 2025.
2. Strand JR. Introduction to Insect Identification and Sampling. BIOB 260: Principals of Biological Diversity. Montana State University. Instructors: Erin Teichroew and Chris Larson. August 2025.
3. Strand JR. Integrated Pest Management. ENSC 110: Land Resources and Environmental Sciences. Montana State University. Instructor: Frank Dougher. October 2025.

Workshops

1. Understanding Wheat Stem Sawfly. 2022 Integrated Pest Management Workshop, "MSU Extension IPM Workshop: Diagnosing Plant Health Problems" Montana State University, Bozeman, MT.

CO-HOSTED ACADEMIC EVENTS

LEPS 2026. LRES Graduate Student Organization Annual Symposium. April 9, 2026. Montana State University, Bozeman, MT.

- Treasurer
- Link to website

LEPS 2025. LRES Graduate Student Organization Annual Symposium. April 7, 2025. Montana State University, Bozeman, MT.

- Chair
- Link to website

2024 LRES-PSPP Student Research Symposium. LRES Graduate Student Organization Annual Symposium. April 18, 2024. Montana State University, Bozeman, MT.

- Chair & event emcee
- Link to website

2023 LRES Colloquium. LRES Graduate Student Organization Annual Symposium. April 13, 2023. Montana State University, Bozeman, MT.

- Treasurer

2022 LRES Colloquium. LRES Graduate Student Organization Annual Symposium. April 29, 2022. Montana State University, Bozeman, MT.

- Treasurer

TEACHING

Instructor of Note

Integrated Pest Management (AGSC 401)
Fall 2026, 3 credits
25 undergraduates, 10 graduate students

Montana State University

Teaching Assistant

Integrated Pest Management (AGSC 401)
Fall 2025, 3 credits, 25 undergraduates, 10 graduate students
Online course taught by Dr. David Weaver
Montana State University

Integrated Pest Management (AGSC 401)
Fall 2023, 3 credits, 20 undergraduates, 10 graduate students
Online course taught by Dr. David Weaver
Montana State University

Invited Guest Lecturer

Principals of Biological Diversity (BIOB 260)
Fall 2026, 5 credits
"Introduction to Insect Identification and Sampling"
Course taught by Dr. Lisa Rew
Montana State University

Role of Plants in the Environment (ENSC 210)
Spring 2026, 3 credits
"Insects for Good: Biological Control"
Course taught by Dr. Will Wetzel
Montana State University

Role of Plants in the Environment (ENSC 210)
Spring 2025, 3 credits
"Can Bugs Help Us?"
Course taught by Dr. Will Wetzel
Montana State University

Principals of Biological Diversity (BIOB 260)
Fall 2025, 5 credits
"Introduction to Insect Identification and Sampling"
Course taught by Erin Teichroew and Chris Larson
Montana State University

Land Resources and Environmental Sciences (ENSC 110)
Fall 2025, 3 credits
"Integrated Pest Management"
Course taught by Frank Dougher
Montana State University

SERVICE

Peer Review Journal Activity

Journal of Environmental Entomology	2024 - present
Journal of Environmental Toxicology and Chemistry	2025 - present

Society Memberships

LRES Graduate Student Organization (Treasurer)	2025 - present
LRES Graduate Student Organization (Co-chair)	2022 - present
LRES Graduate Student Organization (Treasurer)	2021 - 2022
American Association for the Advancement of Science	2026 - present
Entomological Society of America	2021 - present
Montana Grain Growers Association	2022 - present

Active Community Member Volunteer

Southwest Montana Mountain Bike Association	2021 - present
Gallatin Watershed Council	2021 - present
Gallatin Valley Land Trust	2021 - present

EMPLOYMENT HISTORY

Pace Analytical - Onsite at 3M – *Regulatory IT Analyst* 2017 - 2021
(Maplewood, MN)

- Provided digital support between 3M employees and the 3M Chemical Data Management System (CDMS).
- Integrated, tested, and supported user and data interface between CDMS and SAP.
- Generated, designed, and communicated unique, detailed reports of material information using SQL, Microsoft Excel, and SAP.

Metropolitan Mosquito Control District – *Seasonal Mosquito Technician* 2017
(Oakdale, MN)

- Worked on a team to collect and analyze mosquito larvae samples from wetland areas across the St. Paul metropolitan area.
- Independently maintained low adult mosquito populations in Oakdale, MN by surveying and treating both adults and larvae.

Leatherdale Equine Center – *Barn Maintenance Crew* 2015 - 2017
(St. Paul, MN)

- Maintained healthy conditions for equine patients at the University of Minnesota Horse Veterinary Center and Large Animal Hospital.
- Trained and managed new members of the crew.

Minnesota Youth Soccer Association – *Youth Soccer Referee* 2010 - 2021